Complex Analysis

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Contents

1 Complex Analysis (複分析 or 複變分析)

I Derivative

i Derivative

Let W be a topological vector space. The ordinary derivative or derivative of a function $f(u): U \subseteq \mathbb{C} \to W$ (with respect to u) at $x \in U$, denoted as $f'(x)$, is defined as

$$f'(x) = \lim_{z \to x} \frac{f(z) - f(x)}{z - x}$$

if the limit exists. If such limit exists, we say f is (complex) differentiable at x.

We define the (first(-order)) ordinary derivative (function) or derivative (function) of f (with respect to u) as a function f' with codomain W such that for any $x \in U$ at which f is (complex) differentiable, f' maps x to the derivative of f at x.

The derivative of the kth(-order) ($k \in \mathbb{N}$) derivative function of f at $x \in U$ is called the $(k+1)$th(-order) derivative of f (with respect to u) at $x \in U$. The derivative function of the kth(-order) ($k \in \mathbb{N}$) derivative function of f is called the $k+1$th(-order) derivative function of f (with respect to u).

If for any $n \in \mathbb{N}$, the nth(-order) derivative of a function f at a point x in its domain exists, we say f is infinitely (complex) differentiable at x.

If f is (complex) differentiable at all point in $I \subseteq U$, we say f is (complex) differentiable on I; if f is (complex) differentiable on U, we say f is (complex) differentiable. If $f^{(n-1)}$ exists and is (complex) differentiable at all point in $I \subseteq U$, we say f is n-times (complex) differentiable on I; if $f^{(n-1)}$ exists and is (complex) differentiable on U, we say f is n-times (complex) differentiable.

The operation of finding the derivative or derivative function is called ordinary differentiation or differentiation.

Specifically, the 0th(-order) derivative of f is f itself.

ii Smoothness

Same as that of real.

iii Cauchy–Riemann equations

A function $f(x+iy) = u(x,y) + iv(x,y)$ for some real-valued functions u, v with $D_u, D_v \subseteq \mathbb{R}_2$ is holomorphic if u and v satisfy the Cauchy–Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y},$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

The converse is true if f is continuous.

Proof. TODO □

iv Holomorphic or analytic function

A function f is holomorphic or analytic on $U \subseteq D_f$ if it is complex differentiable at every $z \in U$.

A function f is holomorphic or analytic if it is holomorphic on D_f.

A function f is holomorphic or analytic at $z \in D_f$ if it is holomorphic on some open neighborhood of z.

v Biholomorphic function

A function f is biholomorphic on $U \subseteq D_f$ iff f_U is bijective and holomorphic and $(f_U)^{-1}$ is holomorphic.

A function is biholomorphic iff it is bijective and holomorphic and its inverse is holomorphic.

II Poles and Zeros

i Poles

An isolated singularity z_0 of a complex function $f(z)$ is a pole of it iff there exists $n \in \mathbb{N}$, called the order, multiplicity, or degree of z_0, such that

$$(z - z_0)^n f(z)$$

is holomorphic on an open neighborhood of z_0 and

$$\lim_{z \to z_0} \left(z - z_0\right)^n f(z) \neq 0.$$

A simple pole or single pole is a pole of order 1. A multiple pole is a pole of order > 1.

ii Zeros

A point is a zero of a complex function f iff it is a pole of $\frac{1}{f}$.

A point is a zero of order, multiplicity, or degree $-n$ of a complex function f iff it is a pole of order n of $\frac{1}{f}$.

A simple zero or single zero is a zero of order -1. A multiple pole is a pole of order < -1.

iii Meromorphic function

A complex function f is meromorphic on an open set U if for all $z \in U$, either f is holomorphic at z or z is a pole of f.

A complex function f is meromorphic if it is meromorphic on D_f.

iv Pole-zero constellation or plot

A scattering plot on the complex plane showing the poles and zeros of a meromorphic function.

III Essential singularity

An isolated singularity of a complex function is an essential singularity iff it is neither a removable singularity nor a pole.

IV Annulus

An annulus ann$(a; r, R)$ with $R \le \infty$ in the complex plane is a region defined as

$$r < |z - a| < R.$$

If $r = 0$, the region is also called punctured disk.

V Contour

A contour is a parameterized curve $\gamma(t) : [a, b] \to \mathbb{C}$ that is piecewise C^1.

VI Contour integral

The contour integral of a complex function f defined on the image of a contour $\gamma(t) : [a, b] \to \mathbb{C}$ over γ is defined as the Riemann–Stieltjes integral:

$$\int_\gamma f(z)\,\mathrm{d}z := \int_{t=a}^{b} f(\gamma(t))\,\mathrm{d}\gamma(t)\,.$$

i Cauchy integral theorem or Cauchy–Goursat theorem

TODO

ii Cauchy's integral formula

TODO

iii Laurent series

Let f be holomorphic on an annulus $A = \mathrm{ann}\big(z_0; r; R\big)$. The Laurent series of f at z_0 is

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}, \quad z \in A,$$

where the first term is called regular or analytic part and the second term is called principal or singular part, and a_n is given by

$$a_n = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}}\,\mathrm{d}z$$

for any closed contour $\gamma \in A$ enclosing z_0.

- $\forall n \in \mathbb{N} :\ a_{-n} = 0$: either f is holomorphic at z_0 or z_0 is a removable singularity of f.
- $\exists N \in \mathbb{N}$ s.t. $a_{-N} \neq 0 \wedge \forall n \in \mathbb{N}_{>N} :\ a_{-n} = 0$: z_0 is a pole of order N of f.

- Otherwise: z_0 is a essential singularity of f.

a_{-1} is called residue of f at z_0, denoted as $\mathrm{Res}\big(f, z_0\big)$ or $\mathop{\mathrm{Res}}_{z_0}(f)$, that is,

$$\mathrm{Res}\big(f, z_0\big) = \frac{1}{2\pi i} \oint_\gamma f(z)\,\mathrm{d}z\,.$$

iv (Cauchy's) residue theorem

Let U be a simply connected open subset of the complex plane containing a finite list of points $a_1, \ldots, a_n$, $U_0 = U \setminus \{a_1, \ldots, a_n\}$, and a function f is holomorphic on U_0. Let γ be a closed rectifiable curve in U_0. Let the winding number of γ around a_k be $I(\gamma, a_k)$.

$$\oint_\gamma f(z)\,\mathrm{d}z = 2\pi i \sum_{k=1}^{n} I(\gamma, a_k)\,\mathrm{Res}\big(f, a_k\big).$$

Proof. By divergence theorem, obviously. □